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10.2.2 A simple shifted QR algorithm

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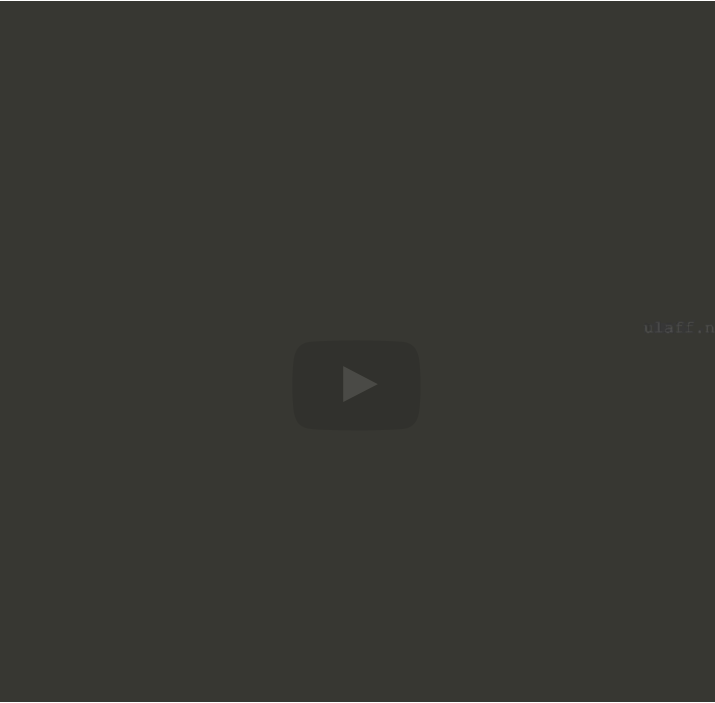
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what that shift is by saying, "Oh, we have an approximation for the eigenvector associated with the smallest eigenvalue and we can then use the Rayleigh Quotient to actually compute that shift, an approximation for λ_{m-1} ." " And then we say, "Well, if we computed $\hat{A}$ along the way, if you think about it carefully, it is the $m-1$ $m-1$ entry in this matrix $\hat{A}$. That is exactly equal to the Rayleigh Quotient

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Reading assignment

1/1 point (graded)

Read Unit 10.2.2 of the notes. [\[LINK\]](#)

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? SimpleShiftedQRAlg.gives different results?	2

1/1 point (graded)

Copy `SimpleQRAlg.m` into `SimpleShiftedQRAlgConstantShift.m` and modify it to implement an algorithm that executes the QR algorithm with a shifted matrix $A - \rho I$:

```
function [ Ak, V ] = SimpleShiftedQRAlgConstantShift( A, rho,
maxits, illustrate, delay )
```

Modify the appropriate line in `Assignments/Week10/matlab/test_simple_QR_algorithms.m`, changing `(0)` to `(1)`, and use it to examine the convergence of the method.

Try different values for ρ : **0.0, 0.9, 0.99, 1.0, 1.99, 1.5**. What do you observe?

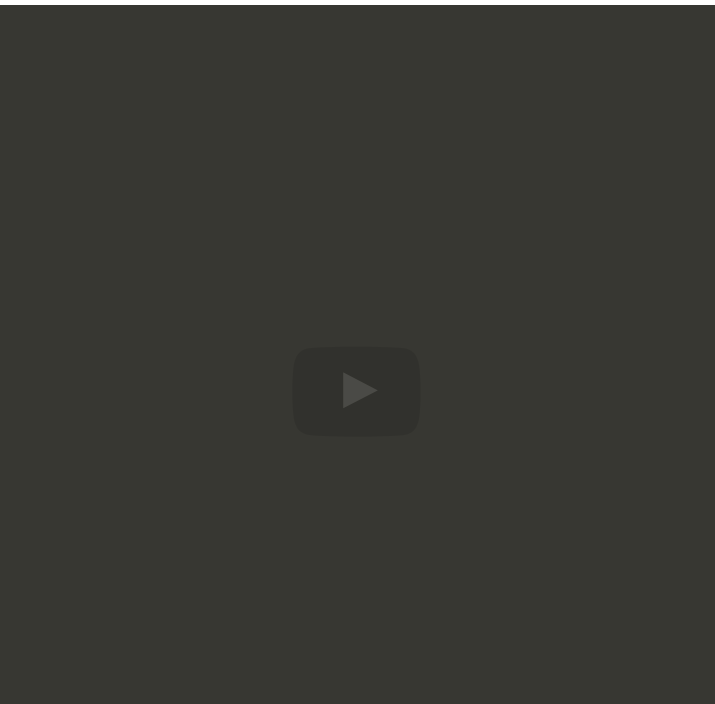
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We end this unit by stating the Shifted QR Algorithm a little bit more explicitly with it more intermediate values exposed so that some of the homeworks that we are going to ask you can refer back to it.

And notice that it explicitly tracks what the shift that was computed at each step.

And for the rest it should be self-explanatory.

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Homework 10.2.2.2

1/1 point (graded)

For the above algorithm, show that

- $A^{(k+1)} = Q^{(k+1)H} A^{(k)} Q^{(k+1)}.$
- $\lambda^{(k+1)} = \tau^T(k+1) H \lambda^{(k)} \tau(k+1)$

☒ done



Solution:

For a complete solution see the [notes](#).

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Answers are displayed within the problem

Homework 10.2.2.3

1/1 point (graded)
For the above algorithm, show that

$$\begin{aligned} & (A - \mu_{k-1}I) (A - \mu_{k-2}I) \cdots (A - \mu_1I) (A - \mu_0I) \\ &= \underbrace{Q^{(0)} Q^{(1)} \cdots Q^{(k)}}_{\text{unitary}} \underbrace{R^{(k)} \cdots R^{(1)} R^{(0)}}_{\text{upper triangular}}. \end{aligned}$$

☒ done



Solution:

For a complete solution see the [notes](#).

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Answers are displayed within the problem

Homework 10.2.2.4

1/1 point (graded)
Copy `SimpleShiftedQRAlgConstantShift.m` into `SimpleShiftedQR.m`
and modify it to implement an algorithm that executes the QR algorithm
in Figure 10.2.2.1:

```
function [ Ak, V ] = SimpleShiftedQRAlg( A, maxits, illustrate,
delay )
```

Modify the appropriate line in
[Assignments/Week10/matlab/test_simple_QR_algorithms.m](#), changing
`(0)` to `(1)`, and use it to examine the convergence of the method.

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